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PAVAN SATYA KRISHNA VENIGALLA
DATA ANALYTICS ENGINEER

PROFILE SUMMARY

- Cloud and Data Platform Engineer with 5+ years of experience designing, automating, and managing enterprise cloud infrastructure, CI/CD pipelines, and scalable data platforms across healthcare, financial services, and enterprise environments.
- Strong hands-on expertise with Microsoft Azure, Azure DevOps, YAML-based CI/CD pipelines, Azure Synapse Analytics, Azure Databricks, ADLS Gen2, Apache Spark, and Delta Lake.
- Experienced in automating deployments across development, staging, and production environments, managing cloud resources, configuring access controls, and implementing infrastructure and deployment best practices.
- Experienced with **ETL orchestration and pipeline scheduling** using Azure Synapse Pipelines, Spark job orchestration, and Azure DevOps, mapping directly to Airflow-style workflow automation patterns.
- Demonstrated ability to **partner with technical and non-technical stakeholders** to translate business and compliance requirements into scalable, production-grade data solutions in fast-paced, agile environments.
- Working knowledge of **AI-assisted development tools** (GitHub Copilot, Amazon Bedrock) and machine learning techniques to accelerate analytical workflows and anomaly detection.

TECHNICAL SKILLS

- **Operating Systems:** Windows, Mac OS, Ubuntu
- **Cloud Platforms:** Microsoft Azure, Azure Synapse Analytics, Azure Databricks, ADLS Gen2, Azure Data Factory, Azure Key Vault, Azure Fabric, AWS, GCP
- **DevOps & CI/CD:** Azure DevOps, Azure Pipelines, YAML Pipelines, CI/CD Automation, Build and Release Pipelines, Git, GitHub, Deployment Automation, Environment Management
- **Infrastructure & Cloud Administration:** Azure Workspace Administration, Spark Pool Management, Storage Account Management, Access Control, RBAC, Service Connections, Linked Services, Key Vault Integration, Cloud Resource Management
- **Programming & Scripting:** Python, PySpark, Bash, PowerShell, SQL, T-SQL, PL/SQL
- **Containers & Distributed Computing:** Apache Spark, Databricks, Delta Lake, Distributed Processing, Containerization
- **Monitoring & Observability:** Azure Monitor, Azure Log Analytics, Structured Logging, Pipeline Monitoring, Execution Auditing, Error Handling, Performance Monitoring
- **Data Architecture:** Medallion Architecture, Lakehouse Architecture, Data Warehousing, ETL/ELT, Dimensional Modeling, Delta Lake Optimization
- **Development Practices:** Agile/Scrum, Version Control, Code Reviews, Release Management, Deployment Troubleshooting, Technical Documentation, Data Governance

WORK EXPERIENCE

Client: Los Angeles County Department of Health Services, Los Angeles, CA, USA **Nov 2025 - Present**
Role: Data Analytics Engineer

- Architected and managed enterprise Azure data platform components supporting the migration of clinical and operational data from Oracle Millennium HealtheIntent to Azure cloud infrastructure.
- Designed, developed, and maintained automated CI/CD pipelines using YAML and Azure DevOps to deploy Synapse notebooks, SQL scripts, and data platform assets across development, staging, and production environments.
- Administered Azure Synapse workspaces, Apache Spark pools, ADLS Gen2 storage containers, access permissions, Key Vault linked services, and Azure DevOps service connections to support secure and reliable deployments.
- Managed deployment configurations, environment-specific parameters, and infrastructure dependencies to streamline release management and maintain consistency across multiple environments.
- Automated data ingestion, transformation, and processing workflows using Python, PySpark, T-SQL, and PL/SQL within Azure Synapse, supporting reliable execution of enterprise-scale workloads.
- Implemented structured logging, execution tracking, error handling, and auditing frameworks within Spark and Python pipelines to improve operational visibility, troubleshooting, and production reliability.
- Optimized Delta Lake workloads through liquid clustering and Z-order indexing, improving query performance, resource utilization, and processing efficiency.
- Established automated validation and UAT frameworks to verify deployment outputs, schema consistency, transformation accuracy, and data integrity across environments.

- Collaborated with engineering, QA, and business teams to troubleshoot deployment issues, resolve infrastructure dependencies, and enforce consistent development and release practices.

I Net Software Technologies Inc, TX, USA **Jun 2025 – Oct 2025**
Role: Data Engineer Intern

- Designed and maintained automated CI/CD workflows using Azure DevOps to support deployment and orchestration of Azure Databricks notebooks, data pipelines, and related platform assets.
- Developed and optimized PySpark and Python-based workflows within Azure Databricks, supporting scalable data processing and automated execution of ETL/ELT workloads.
- Implemented Azure Databricks Lakehouse architecture using Medallion Architecture and Delta Lake, establishing standardized data processing and deployment workflows.
- Configured Unity Catalog for centralized data governance, access management, permission control, and data lineage across Databricks environments.
- Automated recurring data engineering tasks using Azure Data Factory, Databricks Jobs, and Azure DevOps pipelines, improving scheduling consistency and operational reliability.
- Implemented data validation, exception handling, and structured logging mechanisms to identify pipeline failures, monitor execution status, and improve troubleshooting efficiency.

Rutgers University, NJ, USA **Oct 2023 - May 2025**
Role: Data Science Analyst

- Developed and maintained Python-based analytical workflows to process large-scale biological and experimental datasets, supporting reproducible research and computational analysis.
- Automated data preprocessing, transformation, and validation workflows using Python, SQL, and analytical libraries, improving consistency and reliability of research datasets.
- Worked with distributed data processing concepts and computational environments to execute analytical workloads and manage research data efficiently.
- Implemented data validation, exception handling, and quality checks to identify inconsistencies, troubleshoot processing issues, and ensure reliable analytical outputs.
- Developed automated analytical workflows and visualizations to communicate research findings and technical results to cross-functional research teams.
- Collaborated with faculty and research collaborators to document workflows, establish analytical standards, and improve the reliability and reproducibility of research processes.

Piramal Pharma, Bangalore, India **Jun 2021 - Jul 2023**
Role: Data Analyst – Pharma & Finance

- Developed and maintained scalable ETL/ELT workflows using Python, PySpark, SQL, and Databricks to process large-scale healthcare operational and patient datasets.
- Implemented automated data ingestion, transformation, and validation workflows leveraging Apache Spark and Delta Lake to improve processing reliability and data accessibility.
- Optimized Spark-based processing workflows and SQL transformations to improve execution efficiency and support scalable analytical workloads.
- Developed reusable data processing components and automated validation mechanisms to improve consistency, reduce manual intervention, and streamline recurring data operations.
- Implemented data quality monitoring, anomaly detection, and root-cause analysis using Python and SQL to identify processing discrepancies and improve pipeline reliability.
- Collaborated with data engineering, clinical operations, and business teams to troubleshoot data integration issues, support reporting requirements, and maintain reliable data delivery.

EDUCATION

- Masters in **Data Science** from **Rutgers University**, New Jersey, USA
- Bachelors in **Instrumentation and Control Engineering** from **National Institute of Technology, Tiruchirappalli**, India

CERTIFICATIONS

• AWS AI Cloud Practitioner	• Microsoft AZ 400 (On going)
• Microsoft DP 700 (Expired)	• Google Data Analytics Professional